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BANARAS HINDU
UNIVERSITY

AN INSTITUTION OF NATIONAL IMPORTANCE ESTABLISHED BY AN ACT OF PARLIAMENT

A Project on

“Impact of physical activities on academic performance and mental well-being among students”

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In

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Department of Statistics

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CERTIFICATE

This is to certify that the dissertation submitted by Srishti Rao, Roll No. 23220STA133, Enrolment No. 462696, Institute of Science, Banaras Hindu University, Varanasi, titled as **“Impact of physical activities on academic performance and mental well-being among students”** is a record of research work done by her during the academic year 2025-26 under my/our supervision in partial fulfilment for the award of the degree of Three year BSc. (Hons.) Statistics. This dissertation has not been submitted for the award of any degree, diploma, fellowship or other title. It has not been sent for any publication or presentation purpose. I hereby confirm the originality of the work and that there is no plagiarism in any part of the dissertation.

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1. Abstract

Physical activity plays an important role in shaping both the academic success and mental well-being of students. This study explores how regular involvement in physical activities—such as sports, exercise, and recreational movement—affects students’ concentration, memory, and overall academic performance. It also examines the connection between physical activity and mental health, focusing on aspects like stress, anxiety, and emotional stability. The findings suggest that students who engage in regular physical activities tend to perform better academically, as exercise improves blood flow to the brain and enhances cognitive functioning. Additionally, physical activity provides a healthy outlet for stress, helping students feel more relaxed, confident, and motivated. It also encourages better sleep patterns and social interaction, which further support emotional well-being. On the other hand, a lack of physical activity may lead to increased stress, reduced focus, and lower academic achievement. The study highlights the need for educational institutions to promote a balanced lifestyle by integrating physical activities into daily routines. Overall, maintaining an active lifestyle not only supports students’ physical health but also contributes significantly to their academic growth and mental resilience, helping them cope more effectively with the challenges of student life. In conclusion, for students at Banaras Hindu University, incorporating regular physical activity into daily routines can greatly enhance both academic success and mental resilience. Encouraging a balanced lifestyle that integrates physical movement with academic responsibilities can help students cope effectively with academic pressure and maintain overall well-being.

2. Introduction

In today's fast-paced academic environment, students are constantly juggling classes, assignments, exams, and personal responsibilities. This pressure often leads to stress, anxiety, and difficulty in maintaining focus. In the middle of all this, one important aspect that is often overlooked is physical activity. Simple activities like playing sports, going for a walk, practicing yoga, or even light exercise can make a meaningful difference in a student's daily life. They not only keep the body fit but also help the mind stay fresh and active. When students engage in physical activity regularly, they tend to feel more energetic and focused. Exercise improves blood flow to the brain, which helps in better concentration, memory, and understanding. As a result, students are able to perform better in their studies. It also builds discipline and helps in managing time more effectively, which are important skills for academic success. At the same time, physical activity plays a major role in maintaining mental well-being. Student life can sometimes feel overwhelming, but engaging in activities like sports or exercise helps release stress and creates a sense of relaxation. It can improve mood, boost confidence, and provide a healthy break from academic pressure. Activities done with others also create opportunities for social interaction, helping students feel more connected and supported. In a diverse and academically active environment like Banaras Hindu University, students come from different backgrounds and face various challenges. While the university offers many opportunities for physical engagement, not all students are able to make time for them due to their busy schedules. This can sometimes lead to tiredness, lack of focus, and increased stress. Overall, maintaining a balance between studies and physical activity is essential. Even small efforts to stay active can greatly improve both academic performance and mental well-being, helping students lead, healthier and more fulfilling university life.

2.1 Background of the Study

In today's competitive academic environment, students increasingly face challenges related to academic performance and mental well-being. The pressure to achieve high grades and manage coursework often leads to stress and anxiety. At the same time, technological advancements and changing lifestyles have reduced physical activity among students. Many spend long hours studying or using digital devices, leaving little time for exercise.

Physical activity is essential for a healthy lifestyle and plays a key role in maintaining physical and mental health. Activities like walking, running, and sports help improve blood circulation, boost energy, and enhance brain functioning. Students who engage in physical activity often show better concentration, memory, and focus, which positively affects academic performance.

Physical activity is also closely linked to mental well-being. Students commonly experience stress due to academic pressure and competition. Regular exercise helps reduce stress, anxiety, and mental fatigue by promoting relaxation and improving mood. It also supports better sleep patterns, which are important for both mental health and academic efficiency. Physically active students are often more confident and emotionally stable.

Despite these benefits, many students do not participate in regular physical activity due to lack of time, workload, or motivation. This imbalance can negatively affect both their performance and well-being.

Therefore, it is important to study the relationship between physical activity, academic performance, and mental well-being among students to promote healthier lifestyles and better outcomes.

2.2 Statement of the problem

Many students today find it difficult to maintain a healthy balance between their academic responsibilities and physical activity. Heavy coursework, tight schedules, and constant academic pressure often push students toward a sedentary lifestyle, where physical movement is limited or ignored. This lack of activity can negatively affect both their academic performance and mental well-being. Students may experience reduced concentration, lower memory retention, and decreased motivation, which ultimately impacts their learning outcomes.

At the same time, limited awareness about the importance of physical activity further worsens the situation. Many students do not fully understand how regular exercise can help reduce stress, improve mood, and enhance cognitive abilities.

As a result, issues like anxiety, fatigue, and emotional imbalance become more common. Even in institutions like Banaras Hindu University, where facilities for sports and recreation are available, student participation often remains low.

This gap highlights the need to address the problem and encourage a more active and balanced student lifestyle.

2.3 Objective of the study

The following are the major objectives of the study:

- To examine the relationship between physical activity and academic performance among students of Banaras Hindu University.
- To analyze the impact of regular physical activity on students' mental well-being, including stress, anxiety, and emotional balance.
- To assess the level of participation of students in various physical activities such as sports, yoga, and exercise.
- To identify the challenges or barriers that prevent students from engaging in regular physical activity.
- To suggest ways to promote a balanced lifestyle by encouraging physical activity for improving both academic outcomes and mental health.

2.4 Scope of the study

This study focuses on understanding the impact of physical activity on academic performance and mental well-being among students of Banaras Hindu University. It covers various forms of physical activities such as sports, yoga, walking, and gym-based exercises that students may engage in during their daily routine. The study aims to examine how these activities influence key academic factors like concentration, memory, and overall performance, along with mental health aspects such as stress, anxiety, and emotional balance.

The scope is limited to students from different faculties and courses within BHU, considering their diverse academic backgrounds and lifestyles. It also takes into account factors like time management, participation levels, and access to physical activity facilities available on campus.

However, the study does not cover medical or clinical analysis of health conditions, nor does it extend beyond the university setting. Its primary focus is to provide a general understanding of how physical activity contributes to a healthier and more productive student life within the academic environment.

2.5 Significance of the Study

This study is important as it highlights the role of physical activity in improving students' academic performance and mental well-being. In today's academic environment, students often prioritize studies over physical health, which can lead to stress, reduced concentration, and poor overall well-being. This study helps in creating awareness about the importance of maintaining a balance between academic work and physical activity.

The findings of this study can benefit students by encouraging them to adopt regular physical activity as a part of their daily routine. It shows how exercise can improve focus, memory, and learning ability, ultimately leading to better academic results. At the same time, it emphasizes the positive impact of physical activity on reducing stress, anxiety, and mental fatigue.

This study is also useful for teachers and educational institutions, as it provides insights into how promoting sports and physical activities can enhance students' overall development. Schools and colleges can use these findings to design programs that support both academic excellence and student well-being.

Furthermore, the study can help parents understand the importance of encouraging their children to stay physically active instead of focusing only on academics. It promotes a holistic approach to student development, where both mental and physical health are given equal importance.

Overall, this study contributes to a better understanding of how a healthy and active lifestyle can lead to improved academic performance and a more balanced, stress-free student life.

2.6 Study Limitations

This study has several limitations that should be considered while interpreting the results. Firstly, the study is based on a limited sample of students, which may not represent the entire student population. Therefore, the findings cannot be generalized to all students.

Secondly, the data collected is mainly based on self-reported responses. Students may not always provide completely accurate information about their physical activity, academic performance, or mental well-being, which can affect the reliability of the results.

Another limitation is the time constraint. The study was conducted within a short period, which may not capture long-term effects of physical activity on academic performance and mental health.

Additionally, the study focuses only on selected factors such as physical activity, stress levels, and academic performance, while other important factors like diet, family background, and personal habits are not deeply analysed.

Lastly, there may be some bias in responses due to personal opinions or misunderstanding of questions by the respondents.

Despite these limitations, the study still provides useful insights into the relationship between physical activity, academic performance, and mental well-being among students.

3. Literature Review

Various studies have explored the relationship between physical activity, academic performance, and mental well-being among students. Research by Singh et al. (2012) found that regular physical activity has a positive impact on cognitive functions such as concentration, memory, and classroom behaviour, which ultimately enhances academic performance. The study emphasized that physically active students tend to perform better academically compared to inactive students. Similarly, Donnelly et al. (2016) conducted a comprehensive review and concluded that physical activity contributes significantly to improved brain functioning and learning outcomes. Their findings suggest that integrating physical activity into daily routines can support academic success without negatively affecting study time.

In terms of mental well-being, research by Biddle and Asare (2011) highlighted that physical activity plays an important role in reducing symptoms of stress, anxiety, and depression among students. The study showed that students who engage in regular exercise experience better emotional stability and overall psychological health. Moreover, a study by Lubans et al. (2016) found a strong association between physical activity and improved mental well-being, including higher self-esteem and lower levels of psychological distress among adolescents. However, some researchers have pointed out that the relationship is not always strong. Álvarez-Bueno et al. (2017) concluded that while physical activity has a positive effect, the impact on academic performance is generally small to moderate and depends on factors such as intensity, duration, and type of activity.

Overall, the existing literature suggests that physical activity has a positive influence on both academic performance and mental well-being. However, variations in results indicate the need for further research to better understand the extent and nature of this relationship.

4. Methodology

This study adopts a descriptive research design using primary data collected through a structured questionnaire. A convenience sampling technique is applied to select students. Data related to physical activity, academic performance, and mental well-being is analyzed using percentage methods, tables, and graphical tools to identify patterns and relationships among variables.

4.1 Research Design

The present study is based on a **descriptive and analytical research design**. It aims to describe the patterns of physical activity among students and analyse its impact on their academic performance and mental well-being.

The descriptive aspect of the study focuses on collecting detailed information about students' habits, such as exercise duration, participation in sports, stress levels, sleep patterns, and academic behaviour. This helps in presenting a clear picture of the current situation.

The analytical aspect involves examining the relationship between physical activity (independent variable) and academic performance and mental well-being (dependent variables). It helps in identifying trends, patterns, and possible associations between these variables.

The study is **cross-sectional in nature**, as the data is collected from respondents at a single point in time. The research relies on primary data collected through a structured questionnaire, making the design suitable for understanding real-life student behaviour and drawing meaningful conclusions.

Overall, this research design is appropriate for the study as it allows both description and analysis of the impact of physical activity on students' academic and mental outcomes.

4.2 Data collection

The data for this study was collected using the survey method, which is suitable for gathering information directly from students about their habits, opinions, and experiences. A structured questionnaire was designed to collect primary data relevant to the objectives of the study.

The questionnaire consisted of both closed-ended and multiple-choice questions, making it easy for respondents to answer. It included questions related to physical activity (such as duration of exercise and participation in sports), academic performance (self-assessment and study habits), and mental well-being (stress levels, sleep patterns, and feelings).

The survey was conducted among a selected group of students using a convenience sampling technique. The questionnaire was distributed either in offline mode (printed forms) or online mode (Google Forms), depending on accessibility. Clear instructions were provided to ensure that respondents understood the questions properly.

To maintain the accuracy of the data, respondents were requested to answer honestly, and their responses were kept confidential. The collected data was then compiled, organized, and prepared for analysis.

Thus, the survey method proved to be an effective and practical approach for collecting relevant data for the study.

4.3 Sampling Technique

In this study, the convenience sampling technique was used to select the respondents. Under this method, students who were easily accessible and willing to participate were included in the sample. The sample consisted of students from different academic backgrounds, making it easier to collect data within the limited time available. This technique was chosen due to its simplicity, cost-effectiveness, and suitability for academic projects with time constraints. Although convenience sampling does not ensure complete representation of the entire student population, it is useful for identifying general trends and patterns related to physical activity, academic performance, and mental well-being. Thus, the convenience sampling technique was appropriate for this study as it allowed quick and efficient data collection.

4.4 Tools and Techniques of Analysis

The data collected for this study was analysed using basic statistical tools and techniques to ensure clear understanding and proper interpretation of results. The primary tool used was percentage analysis, which helped in converting raw data into meaningful proportions. This made it easier to compare responses and identify trends among students regarding physical activity, academic performance, and mental well-being. The data was also presented using tabular representation, which organized the information in a systematic and simple format. This helped in better classification and easy reference during analysis. For visual representation, bar graphs and pie charts were used. These graphical tools made the data more understandable and helped in highlighting patterns and comparisons effectively. Additionally, simple comparative analysis was performed to observe differences between physically active and less active students in terms of their academic performance and mental well-being.

Overall, these tools and techniques were suitable for summarizing the data, identifying relationships, and drawing meaningful conclusions from the study.

4.5 Ethical Considerations

Ethical principles were carefully followed during the conduct of this study to ensure fairness, respect, and integrity in data collection and analysis.

Firstly, informed consent was obtained from all participants before collecting data. Students were clearly informed about the purpose of the study and were assured that their participation was completely voluntary.

Secondly, the confidentiality and anonymity of the respondents were maintained. No personal information such as names or identities was disclosed, and the data collected was used only for academic purposes.

Participants were given the freedom to withdraw from the survey at any stage without any pressure or obligation. This ensured that their rights and comfort were respected throughout the process.

Additionally, the questions included in the questionnaire were designed to be non-invasive and appropriate, avoiding any sensitive or personal topics that could make respondents uncomfortable. Lastly, the data collected was reported honestly and accurately, without any manipulation or bias, ensuring the reliability and validity of the study.

These ethical considerations helped in maintaining the credibility of the research and protecting the rights of the participants.

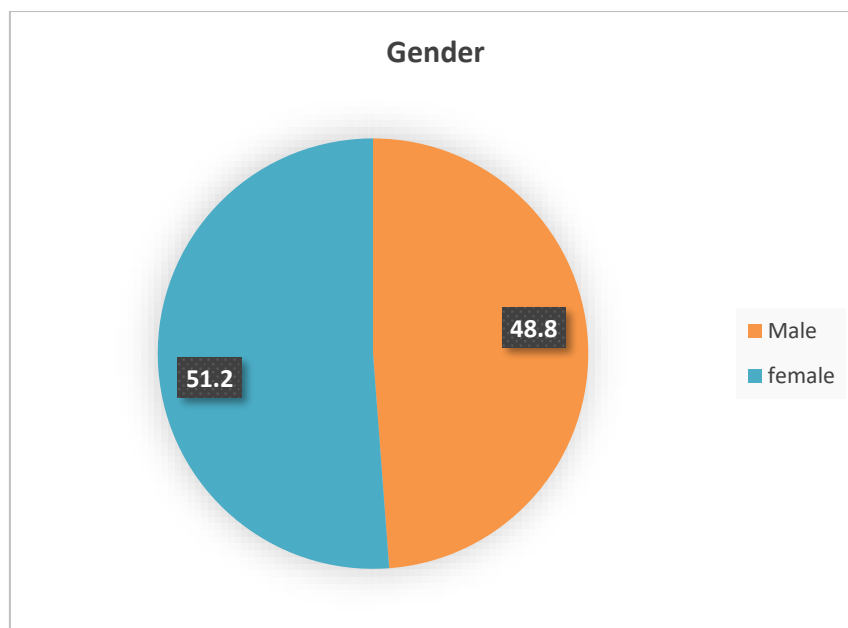
5. Descriptive analysis and interpretation

The descriptive analysis indicates that a significant proportion of students participate in physical activity, though regular engagement remains moderate. Students who are physically active show better concentration, improved academic performance, and lower stress levels. A clear trend is observed where moderate physical activity is associated with better sleep patterns and positive mental well-being. In contrast, students with low physical activity report higher stress, fatigue, and reduced academic efficiency. The graphical analysis supports a positive association between physical activity and both academic and mental outcomes. Overall, the findings highlight that increased physical activity contributes to improved student performance and psychological well-being.

Table1: Distribution of students on the basis of their gender.

GENDER	FREQUENCY	PERCENTAGE (%)
Male	81	48.8
Female	85	51.2
Total	166	100

Pie – chart

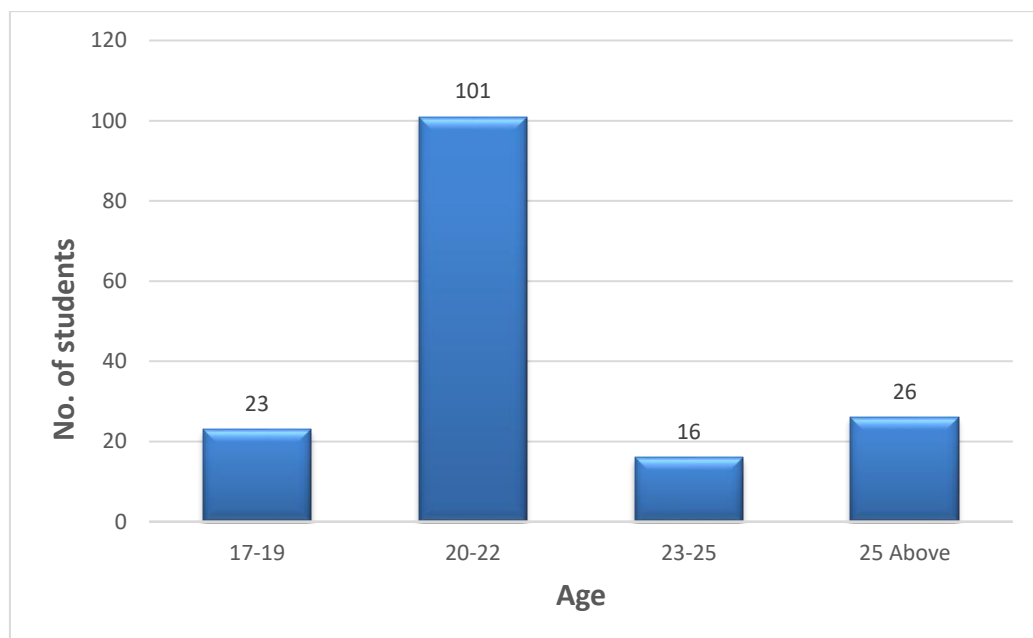


Interpretation: From above we see that the sample is almost evenly distributed, with a slight predominance of female respondents. Such a balanced representation ensures that the findings of the study are unbiased and reflect perspectives from both genders fairly.

Table 2: Distribution of students on the basis of their age.

Age	Frequency	Percentage
17-19	23	13.9
20-22	101	60.8
23-25	16	9.6
25 Above	26	15.7

Bar Chart:

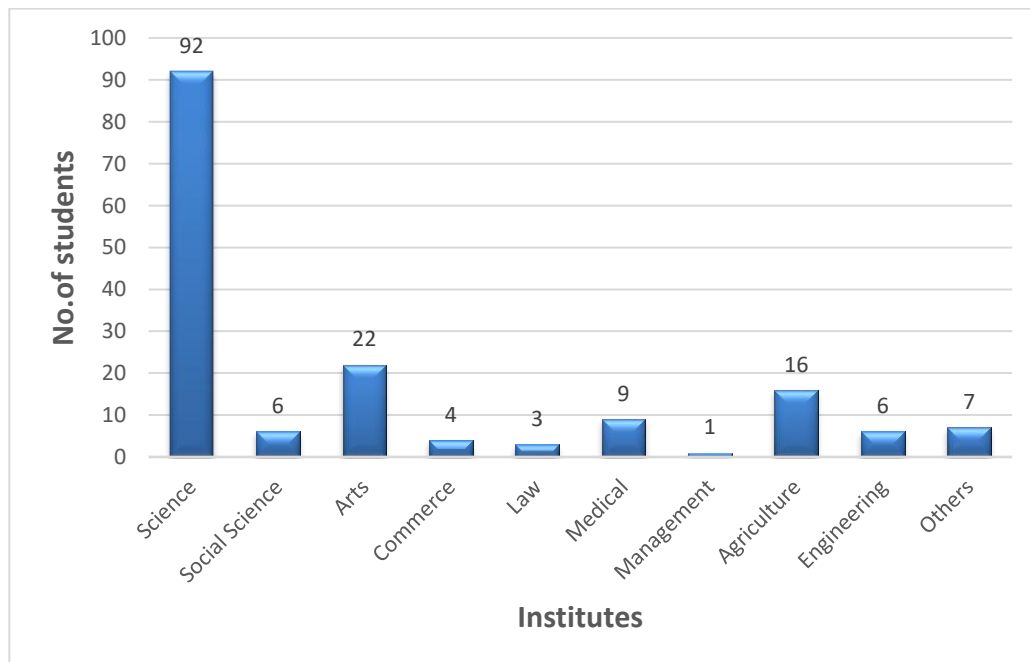


Interpretation: From the above we see that most respondents belong to the 20-22 age group, showing the highest participation. While other age groups have moderate representation.

Table 3: Distribution of students on the basis of their institute.

Institute	Frequency	Percentage
Science	92	55.4
Social science	6	3.6
Arts	22	13.3
Commerce	4	2.4
Law	3	1.8
Medical	9	5.4
Management	1	0.6
Agriculture	16	9.6
Engineering	6	3.6
Others	7	4.2

Bar chart:

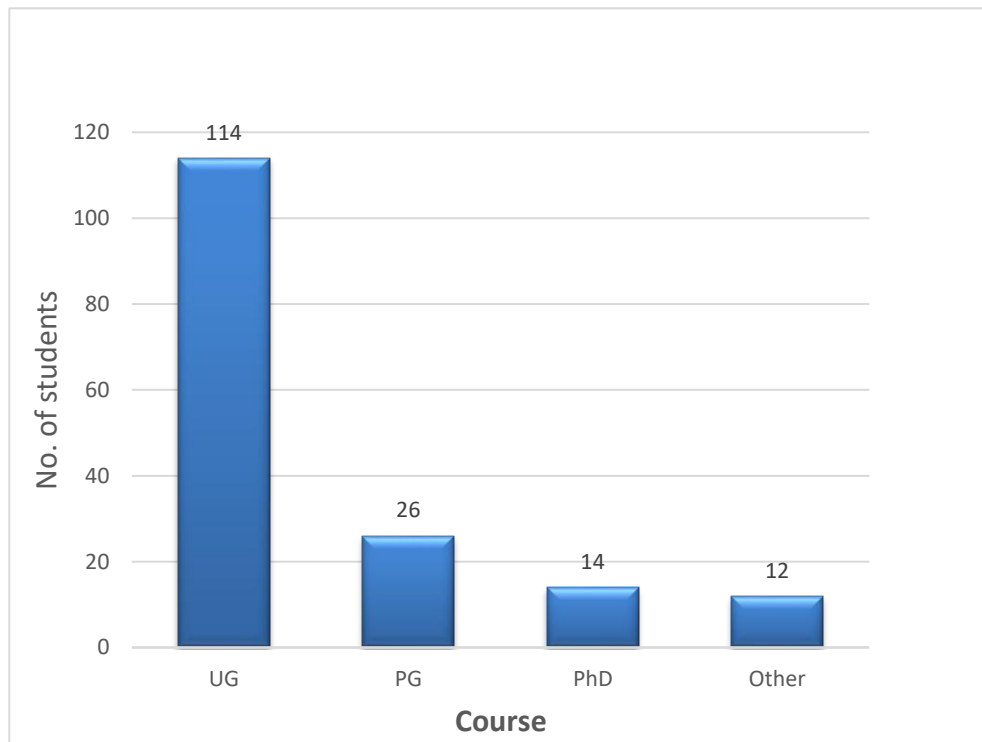


Interpretation: From the above table we see that the sample is mainly dominated by science students, with limited participation from other fields.

Table 4: Distribution of students on the basis of their course.

Course	Frequency	Percentage
Undergraduate	114	68.7
Postgraduate	26	15.7
PhD	14	8.4
Others	12	7.2

Bar chart:

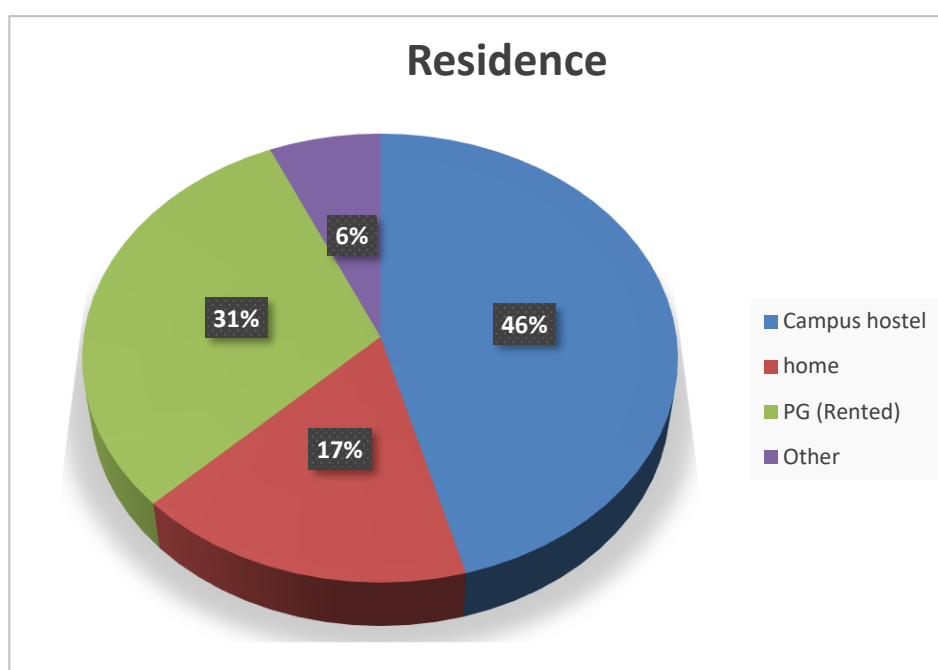


Interpretation: From above we see that the findings of the study are primarily influenced by undergraduate students, with comparatively lower representation from higher academic levels.

Table 5: Distribution of students on the basis of their residence.

Residence	Frequency	percentage
Campus Hostel	76	45.8%
Home	28	16.9%
PG (Rented)	51	30.7%
other	11	6.6%

Pie – chart

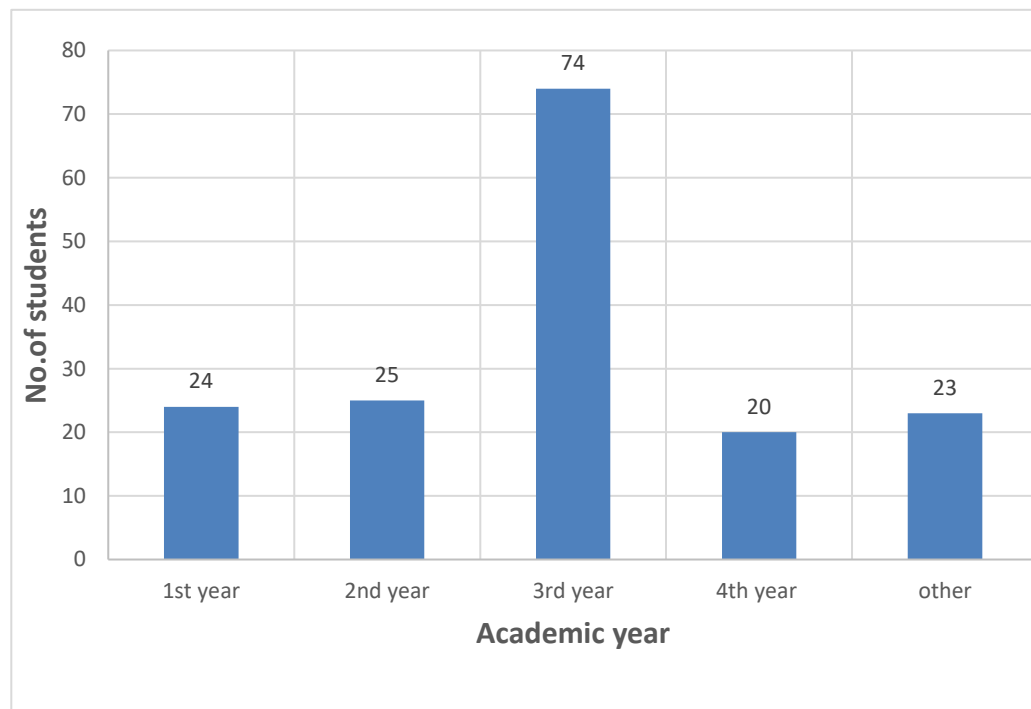


Interpretation: The data shows that the largest proportion of students (45.8%) live in campus hostels, followed by 30.7% in PG accommodations. A smaller percentage lives at home (16.9%), while only 6.6% fall under other categories. Residential environment away from home plays an important role in their lifestyle and performance.

Table 6: Distribution of students on the basis of their academic year.

Academic Year	Frequency	percentage
1 st year	24	14.5%
2 nd year	25	15.1%
3 rd year	74	44.6%
4 th year	20	12%
other	23	13.9%

Bar chart:

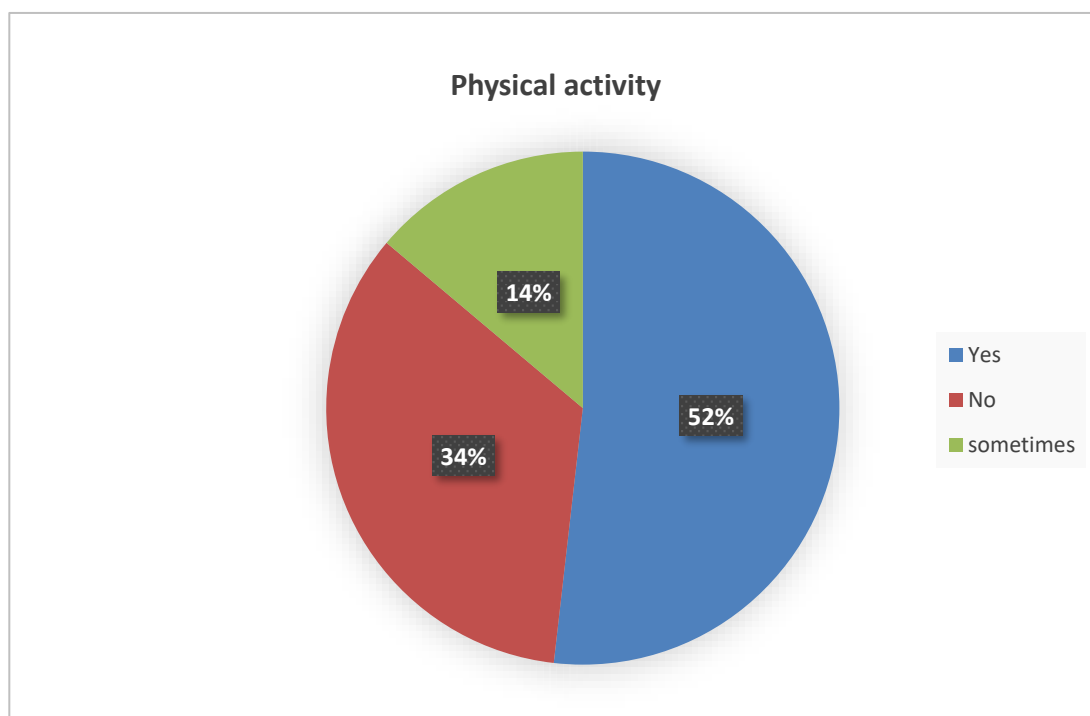


Interpretation: The charts shows the distribution of respondents by academic year. The 3rd year students had the highest participation at approximately 71, while 4th year students had the least at around 19. first and second year students were nearly equal at 23-24 each, with 21 respondents falling under the other category.

Table 7: Distribution of students based on whether they perform physical activity.

Physical activity	Students	percentage
Yes	86	51.8%
No	57	34.3%
Sometimes	23	13.9%

Pie chart:

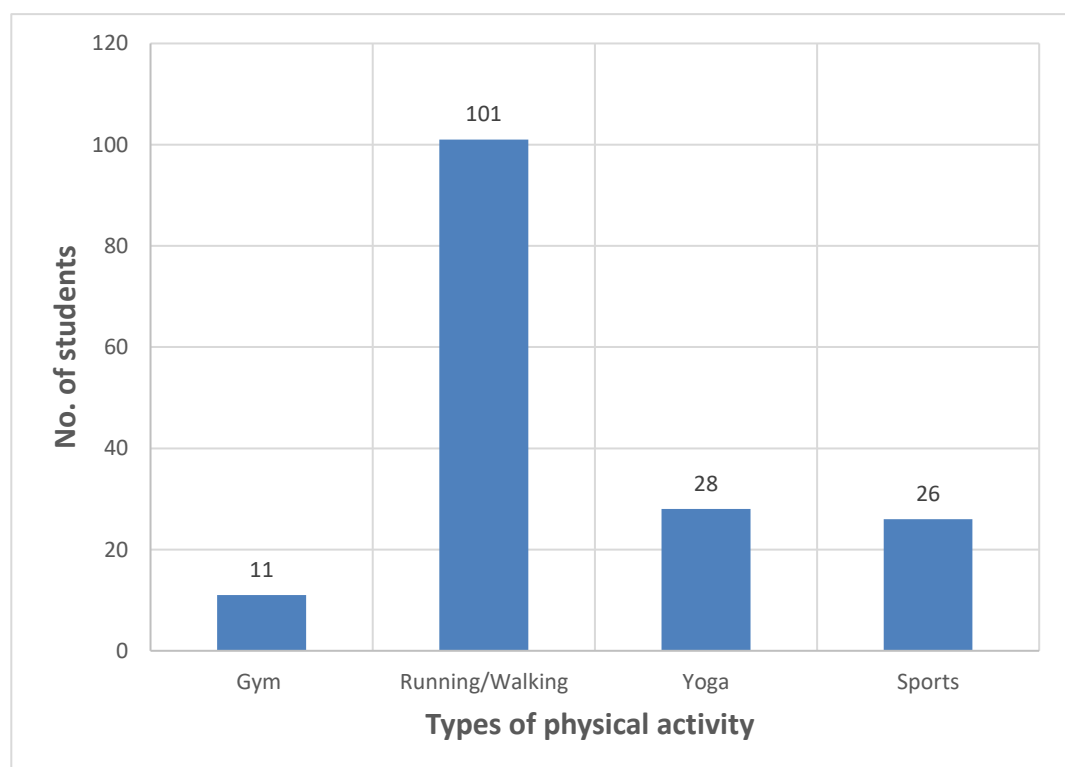


Interpretation: The data shows the distribution of students based on their physical activity habits. It indicates whether students are regularly active, inactive or only occasionally involved in physical activities.

Table 8: Distribution of students based on the type of physical activity they perform.

Type of physical activity	Students	Percentage
Gym	11	6.6%
Running/Walking	101	60.8%
Yoga	28	16.9%
Sports	26	15.7%

Bar chart:

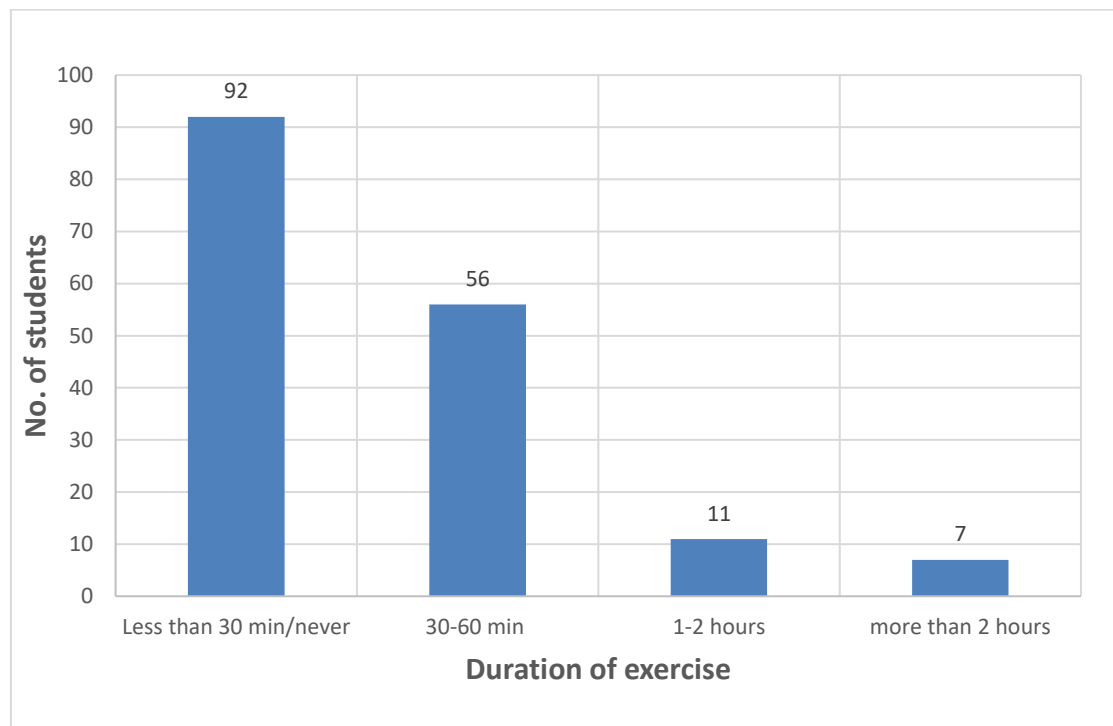


Interpretation: The above chart shows that most students prefer simple activities like walking over intense exercises. Only a smaller group regularly engages in activities like gym or sports.

Table 9: Distribution of students on the basis of duration of daily exercise.

Duration of exercise	students	percentage
Less than 30 min/never	92	55.4
30-60 min	56	33.7
1-2 hours	11	6.6
More 2 hours	7	4.2

Bar chart:

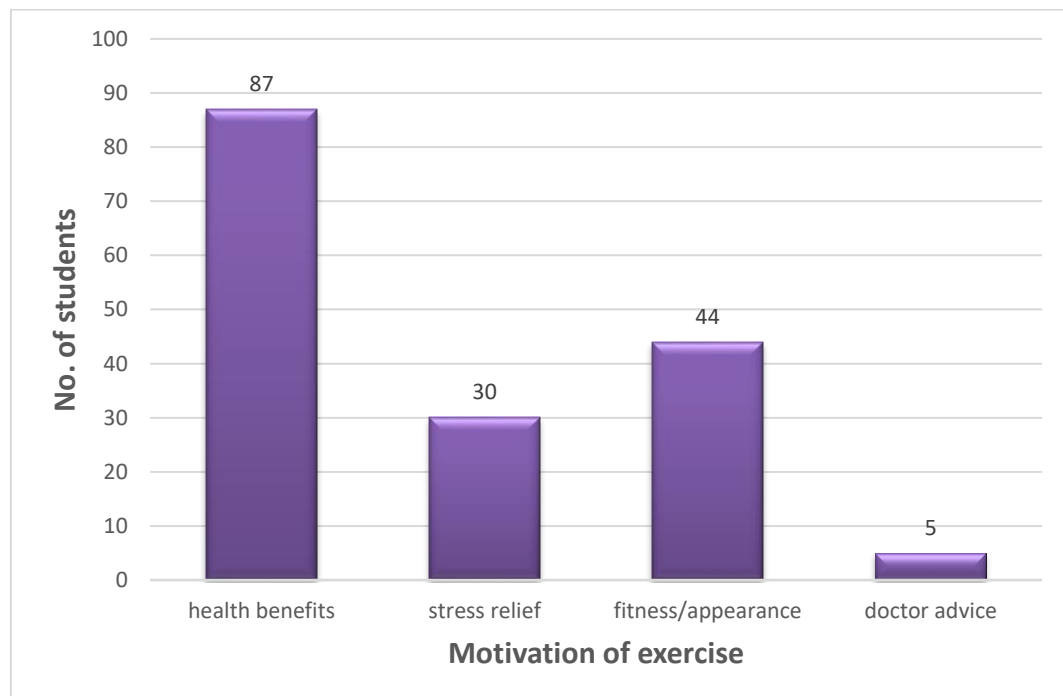


Interpretation: From above data we can see that the majority of students exercise for less than 30 minutes daily, followed by those who exercise for 30-60 minutes. A smaller proportion spend 1-2 hours, while very few students exercise for more than 2 hours. This indicates that most students engage in low to moderate levels of physical activity, with limited participation in longer duration exercise.

Table 10: Distribution of students on the basis of motivation for exercise.

Motivation for exercise	No. of students	percentage
Health benefits	87	52.4%
Stress relief	30	18.1%
Fitness/Appearance	44	26.5%
Doctor/Family advice	5	3 %

Bar chart:

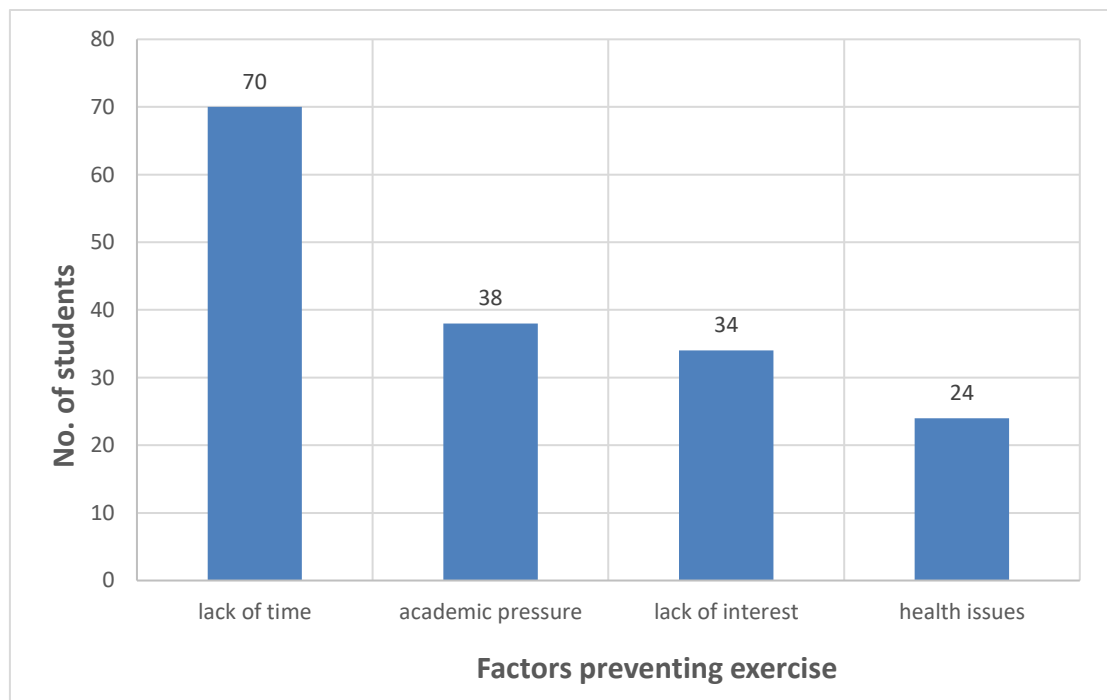


Interpretation: The majority of students are motivated to exercise for health benefits, followed by those who aim to maintain physical fitness. A smaller proportion exercise for stress relief, while very few are motivated by other reasons. This indicates that health-related concerns are the primarily driving factor behind students' participation in physical activity.

Table 11: Distribution of Students on the basis of factors preventing regular exercise.

Factors preventing exercise	No. of students	percentage
Lack of time	70	42.2%
Academic pressure	38	22.9%
Lack of interest	34	20.5%
Health issues	24	14.5%

Bar chart:

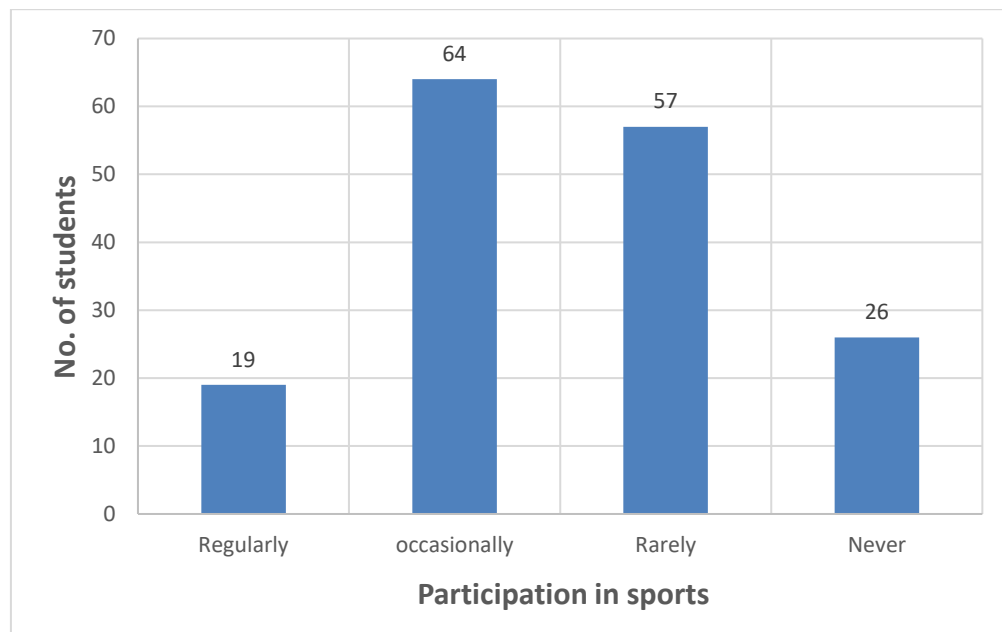


Interpretation: From above chart we can see that most students feel that they don't have enough time to exercise regularly, mainly due to their busy daily schedules. Academic pressure also plays a major role in limiting their physical activity. Some students lack interest in exercising, while only a few are unable to do so because of health issues.

Table 12: Distribution of students on the basis of participation in sports activities.

Participation in sports	No. of students	percentage
regularly	19	11.4
Occasionally	64	38.6
Rarely	57	34.3
Never	26	15.7

Bar chart:

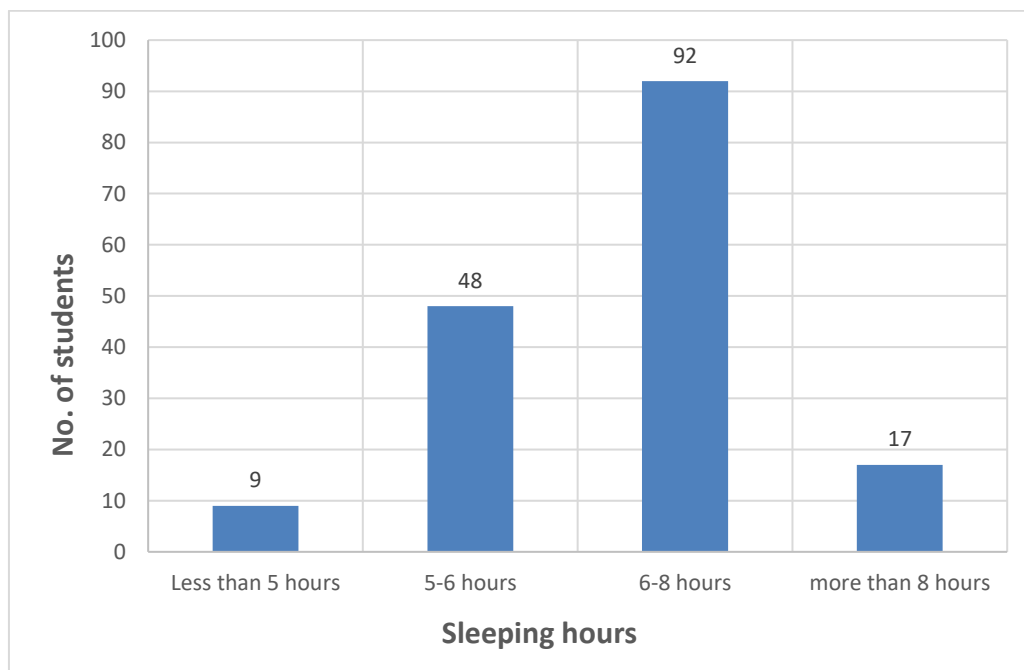


Interpretation: The data shows the majority of students participate in sports activities only occasionally, indicating irregular involvement. This is followed by students who rarely take part, while fewer students never participate. Only a small number of students are regularly involved in sports. This suggests that consistent participation in sports is low, and most students engage only from time to time.

Table 13: Distribution of students on the basis of their daily sleeping hours.

Sleeping hours	No. of students	percentage
Less than 5 hours	9	5.4
5-6 hours	48	28.9
6-8 hours	92	55.4
More than 8 hours	17	10.2

Bar chart:

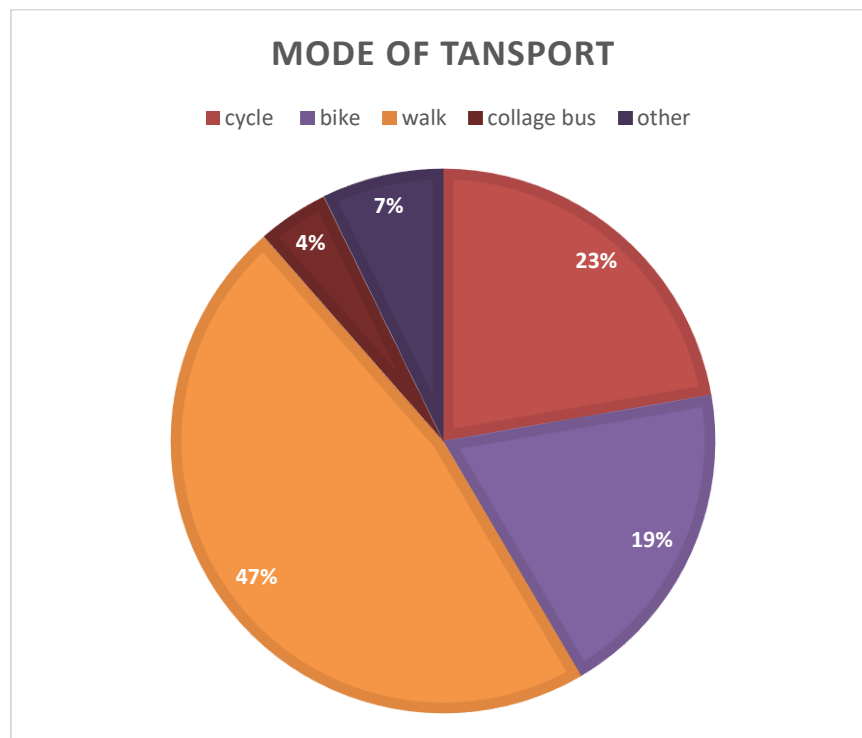


Interpretation: From above chart we can see that the majority of students sleep for 6-8 hours daily, indicating that most maintain a healthy sleep routine. This is followed by students who sleep for 5- 6 hours, suggesting slightly reduced sleep duration. Fewer students sleep for more than 8 hours, while very few get less than 5 hours of sleep. Overall, most students have a fairly balanced sleep pattern, through some may not be getting adequate rest.

Table 14: Distribution of students on the basis of their mode of transport to the collage.

Mode of transport	No. of students	percentage
cycle	37	22.3%
Bike	32	19.3%
walk	78	4.7%
College bus	7	4.2%
Other	12	7.2%

Pie chart:

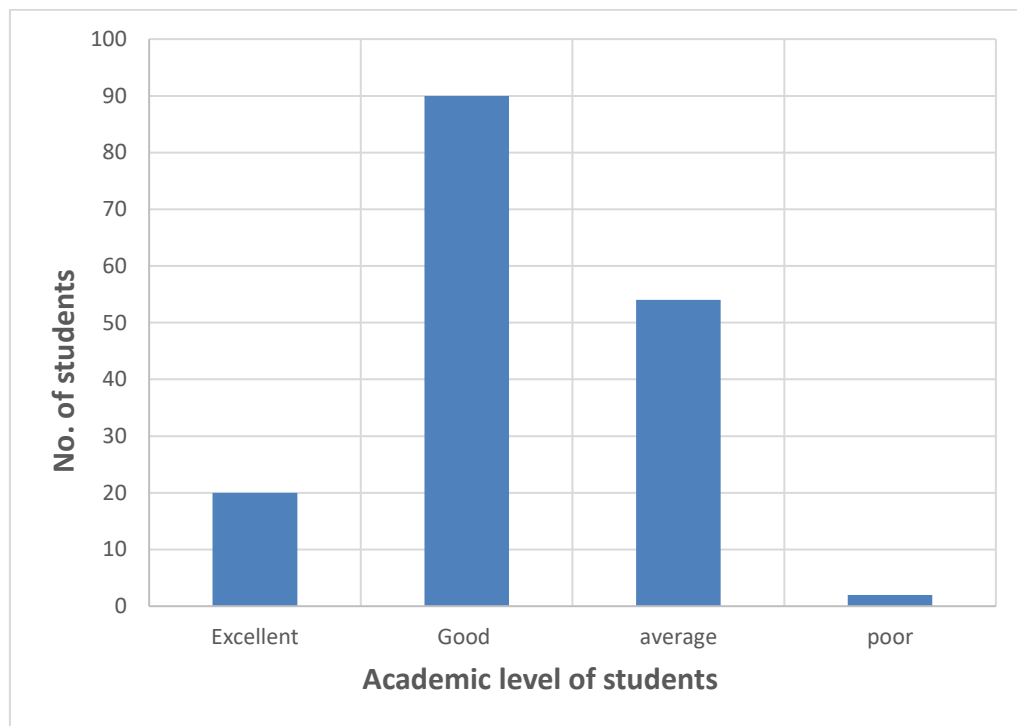


Interpretation: The data shows that the majority of students prefer walking as their mode of transport to collage. This is followed by students using bicycles, indicating a preference for economical and eco-friendly options. A smaller number of students use bikes, while even fewer rely on buses. The least preferred mode fall under other categories.

Table 15: Distribution of students on the basis of their rating of academic levels.

Academic level of students	No. of students	Percentage
Excellent	20	12%
Good	90	54.2%
Average	54	32.5%
Poor	2	1.2%

Bar chart:

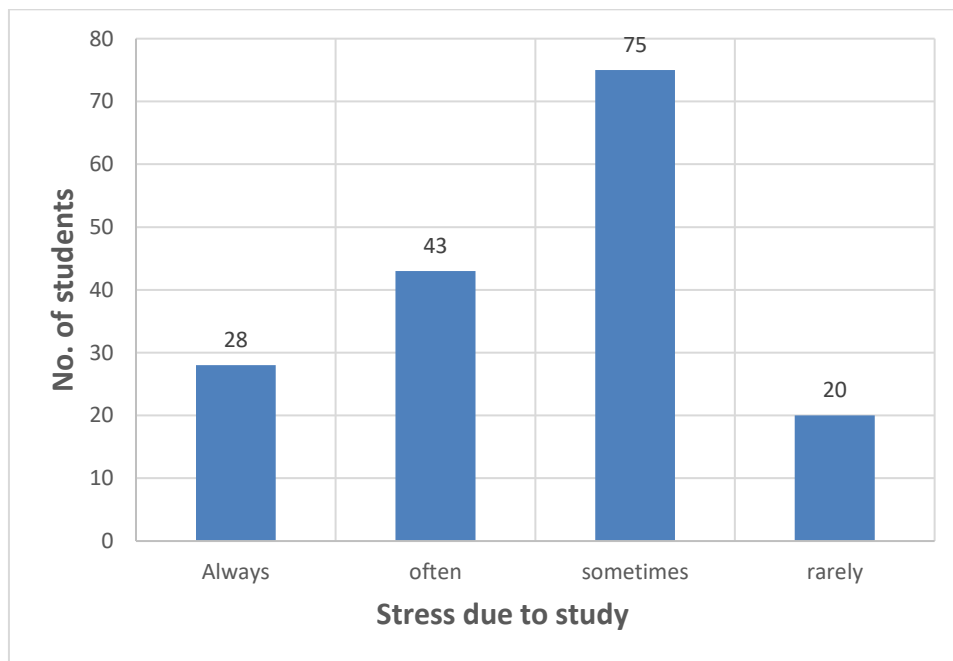


Interpretation: The distribution shows that most students rated their academic level as good, followed by average performance. A smaller proportion of students considered themselves excellent, while very few rated their performance as poor. This indicates that overall academic performance among students is moderate, with the majority falling in the satisfactory range rather than at extreme levels.

Table 16: Distribution of students on the basis of their stress level due to study.

Stress due to study	No. of students	Percentage
Always	28	16.9%
often	43	25.9%
sometimes	75	45.2%
Rarely	20	12%

Bar chart:

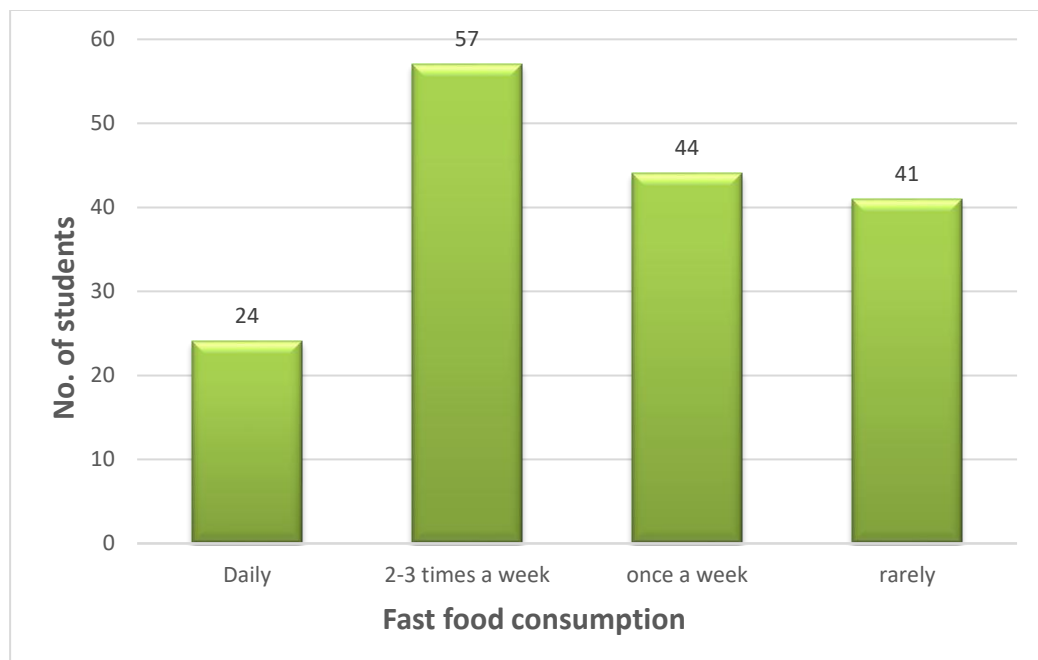


Interpretation: The data indicates that most students experience study-related stress sometimes, followed by those who often feel stressed. A smaller proportion of students report always feeling stress, while very few rarely experience it. This suggests that stress is common among students, but it is generally occasional rather than constant or extremely low.

Table 17: Distribution of students on the basis of their fast food consumption habits.

Fast food consumption	No. of students	Percentage
Daily	24	14.5%
2-3 times a week	57	34.3%
Once a week	44	26.5%
Rarely	41	24.7%

Bar chart:

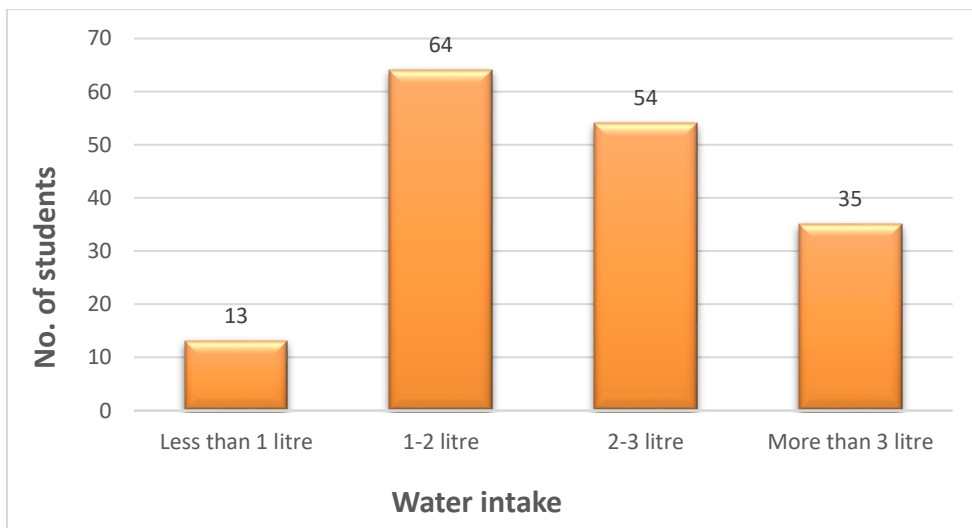


Interpretation: The data shows that most students consume fast food two to three times a week, followed by those who eat it once a week. Fewer students consume fast food rarely, while daily consumption is the least common. This indicates moderate fast food consumption among students rather than frequent daily intake.

Table 18: Distribution of students on the basis of their daily water intake.

Water intake	No. of students	Percentage
Less than 1 litre	13	7.8%
1-2 litre	64	38.6%
2-3 litre	54	32.5%
More than 3 litre	35	21.1%

Bar chart:

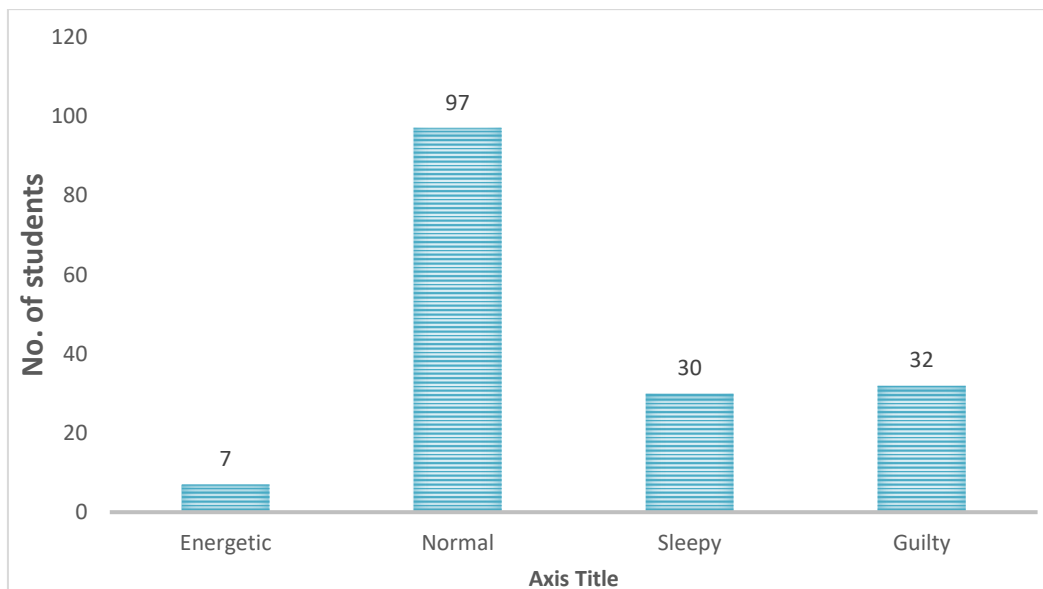


Interpretation: The above chart shows that how much water students drink daily. Most students prefer drinking 1-2 litres of water (38.6%), making it the most common habit among them. A large number of students (32.5%) also drink 2-3 litres, which reflects a fairly good level of hydration. Meanwhile, 21.1% of students consume more than 3 litres , showing that some students are highly conscious about staying hydration. However, a small group (7.8%) drinks less than 1 litre, which is quite low and may not be sufficient for maintaining good health.

Table 19: Distribution of students based on their feelings after consuming fast food.

Feeling of students	No. of students	Percentage
Energetic	7	4.2%
Normal	97	58.4%
Sleepy	30	18.1%
Guilty	32	19.3%

Bar chart:

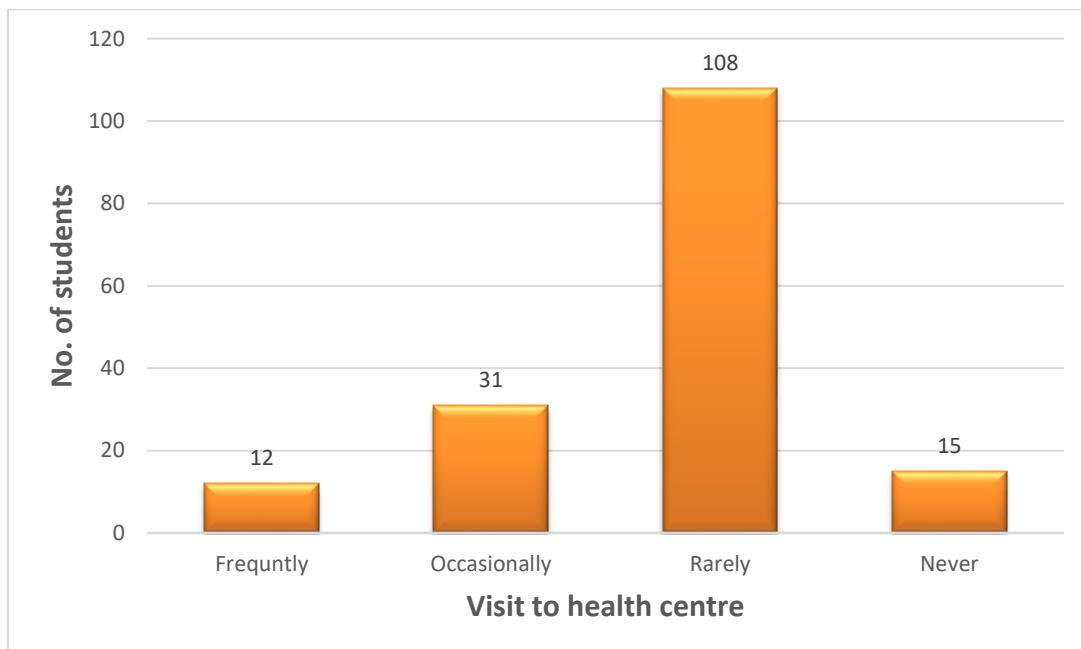


Interpretation: The table shows how students feel after consuming fast food. A majority of students (58.4%) reported feeling normal, indicating that fast food does not cause any immediate noticeable effect for most of them. However, a significant number of students experience negative effects-19.3% feel guilty and 18.1% feel sleepy after eating fast food, suggesting some level of emotional discomfort. Only a small proportion (4.2%) feel energetic, showing that fast food rarely boosts energy levels.

Table 20: Distribution of students based on their visits to the health center.

Visit to health center	No. of students	Percentage
Frequently	12	7.2%
Occasionally	31	18.7%
Rarely	108	65.1%
Never	15	9%

Bar chart:

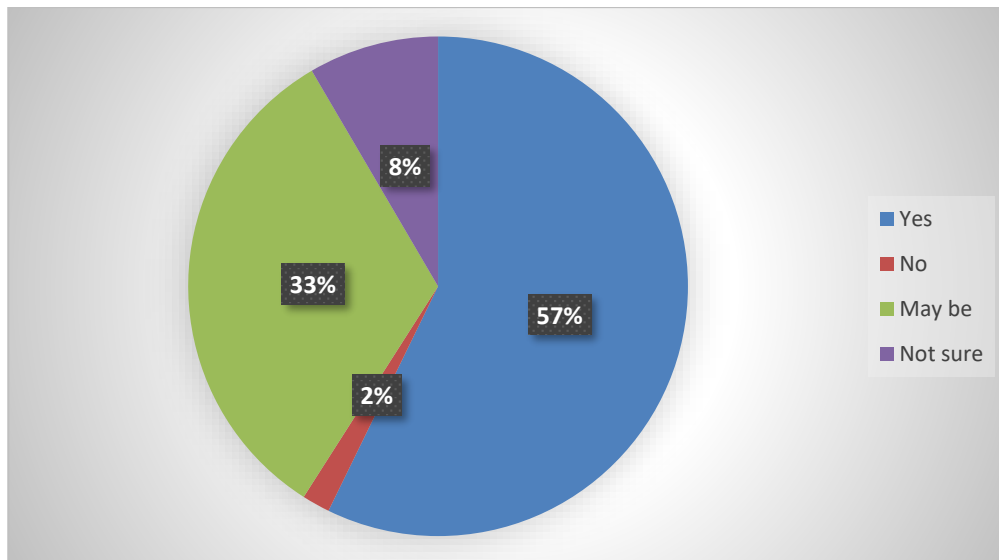


Interpretation center. A large majority of students (65.1%) rarely visit the health center, making it the most common response. The table shows how often students visit the health. This is followed by 18.7% who visit occasionally, indicating that some students seek medical help when needed but not regularly. A smaller proportion of students, 9% never visit the health center at all. Only 7.2% of students visit frequently, which is the lowest category.

Table 21: Distribution of students based on their opinion that active students perform better academically.

Opinion	No. of students	Percentage
Yes	95	57.2%
No	3	1.8%
May be	54	32.5%
Not sure	14	8.4%

Pie chart:



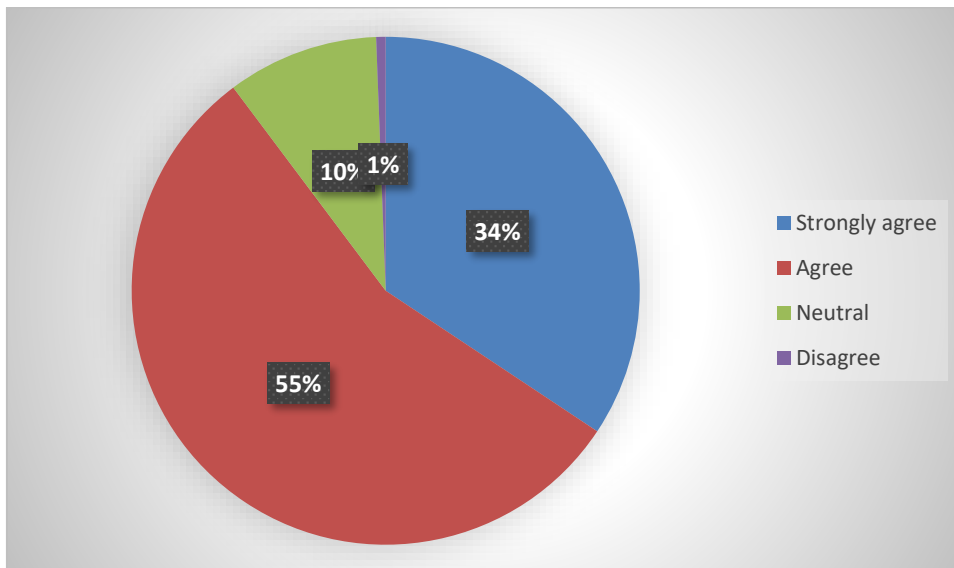
Interpretation: The data shows that most students (57.2%) believe that active students perform better academically. A good number of students (32.5%) are unsure and answered “may be”. Very few students (1.8%) do not agree with this statement, while (8.4%) are not sure.

Overall, it can be said that the majority of students think physical activity has a positive effect on academic performance.

Table 22: Distribution of students based on their opinion that physical activity improves students academic and mental well-being.

Opinion	No. of students	Percentage
Strongly agree	57	34.3%
Agree	92	55.4%
Neutral	16	9.6%
Disagree	1	0.6%

Pie chart:



Interpretation: The data shows that a large majority of students believe that physical activity improves academic performance and mental well-being.

STATISTICAL ANALYSIS

This section presents the results of inferential statistical analysis conducted on the collected survey data related to students' health and academic performance. All tests were performed at a 5% level of significance ($\alpha = 0.05$). The null hypothesis (H_0) is rejected when the p-value is less than 0.05.

6.1 What is a Chi-Square (χ^2) Statistic?

A chi-square(χ^2) statistics is a statistical test used to compare observed data with expected data to determine whether there is a significant difference between variables. It helps in understanding whether deviations occur due to chance or due to an actual relationship.

It is mainly used for categorical data, such as:

- Exercise habits
- Eating patterns
- Sleep duration
- Stress levels

The larger the difference between observed and expected values, the larger the chi-square value.

Definition

The chi-square test measures how much the observed frequencies differ from expected frequencies under a given hypothesis.

Key Points:

- Used for categorical data
- Compares observed (O) and expected (E) values
- Depends on sample size and degrees of freedom

Used for:

- Testing independence of variables
- Testing goodness of fit

Formula for Chi-Square

$$\chi^2 = \sum (O_i - E_i)^2 / E_i$$

Where:

- O = Observed frequency
- E = Expected frequency
- i = Category index

Types of Chi-Square Tests

1. Chi-Square Test of Independence

- The chi-square test of independence is used to determine whether two categorical variables are statistically independent or associated.
- It tests the null hypothesis that no relationship exists between the variables.

Example:

A study is conducted to examine whether gender and course choice are related.

Observed frequencies are collected and compared with expected frequencies.

If a significant difference is found, the null hypothesis of independence is rejected, indicating an association between variables.

- If no relationship → variables are independent
- If relationship exists → variables are dependent

Conditions for Application

- Data must be from a random sample
- Variables must be categorical
- Observations must be independent
- Expected frequency ≥ 5

Uses of Chi-square test of independence

- To examine association between two categorical variables
- To analyse contingency tables
- To test whether variables are independent or dependent

2. Chi-square test of goodness of fit

- The chi-square goodness of fit test is used to examine whether the observed frequency distribution of a sample differs significantly from a theoretical (expected) distribution.
- It tests the null hypothesis that the sample data follows the specified distribution.

Conditions for application

- Data should be obtained from a random sample
- Variables must be categorical in nature
- Observations should be independent
- Expected frequencies should be adequately large (generally ≥ 5)

Uses:

- To test the goodness of fit of a distribution.
- To compare observed and expected frequencies.
- To determine whether deviations are due to sampling fluctuation or real difference.

Concept of pooling in Chi-square test

When at least 20% of the cells of expected frequencies are < 5 then we need pooling i.e. we have to combine two rows or column of the observed frequencies as required.

Row pooling: We combine two rows for pooling.

Column pooling: We combine two rows for pooling .

Analysis – 1: Chi-Square Test of Independence

Physical Activity vs Stress level

This test examines the relationship between students' physical activity and their Stress level.

Hypothesis

- H_0 (Null hypothesis): No significant association between physical activity and stress level.
- H_1 (Alternative hypothesis): There is a relationship between physical activity and stress level.

Observed frequency

Physical activity /stress level	Always	Often	Sometimes	Rarely	Total
Regular (R1)	2	4	9	4	19
Occasionally (R2)	10	18	29	7	64
Rarely (R3)	11	16	26	4	57
Never (R4)	5	5	11	5	26
Total	28	43	75	20	166

Observed frequency (Pooled):

Physical activity/strees level	High stress (always+often)	Low/ Moderate stress(sometimes+rarely)	Row total
Physically active (R1+R2)	34	49	83
Physically inactive(R3+R4)	37	46	83
Column total	71	95	166

Expected frequency:

Physical activity/ stress level	High stress	Low / moderate stress
Active	35.49	47.51
Inactive	35.51	47.49

$$\chi^2(\text{cal}) = 0.2185577$$

$$\chi^2(\text{tab}) = 3.841$$

Result

$$\chi^2 (\text{calculated}) < \chi^2 (\text{tabulated})$$

Decision

We fail to reject H_0 at 5% level of significance.

Interpretation:

There is a statistically significant association between physical activity and mental health.

Students who exercise regularly show better mental health.

- Students who rarely exercise report higher stress and poor well-being.
- This indicates that physical activity positively impacts mental health.

Analysis – 2: Chi-Square Test of Independence**Fast Food Consumption vs Health Condition**

This test is conducted to examine whether there is a significant association between students' fast food consumption habits and their health condition.

Hypotheses

- H_0 : There is no significance association between fast food consumption and health condition.
 H_1 : There is a significant association between fast food consumption and health condition.

Observed frequency

Fast food/ visiting health center	Frequently (C1)	Occas ionaly (C3)	Rarely (C4)	Never (C2)	Total
2-3 times a week (R1)	6	11	39	1	57
Daily(R2)	0	5	12	7	24
Once a week (R3)	3	11	27	3	44
Rarely (R4)	3	4	30	4	41
Total	12	31	108	15	166

Observed (pooled) frequency

Fast food/health condition	frequently	occasionally	Rarely	Never	Total
(R1+R2)	6	16	51	8	81
(R3+R4)	6	15	57	7	85
Total	12	31	108	15	166

Expected frequency: $E = (\text{Row total} \times \text{column total}) / \text{Grand total}$

Fast food /health condition	frequently	occasionally	Rarely	Never	Total
(R1+R2)	5.86	15.13	52.7	7.32	81
(R3+R4)	6.14	15.87	55.3	7.68	85
Total	12	31	108	15	166

$$\chi^2(\text{cal}) = 0.3361$$

$$\chi^2(\text{tab}) = 7.815$$

Result:

$$\chi^2 (\text{calculated}) < \chi^2 (\text{tabulated})$$

Decision:

We fail to reject null hypothesis at 5% level of significance.

Interpretation:

There is no significant relationship between fast food consumption and health condition.

- The chi-square value of 0.3361 is far below the critical value of 7.815.
- The p-value is greater than 0.05.
- Therefore, fast food consumption frequency does not significantly affect how often students visit the health center based on the data.

Analysis – 3: Chi-Square Test of Independence**Sleep Duration vs Academic Performance**

This test examines how sleep affects academic performance.

Hypotheses

- H_0 : No association between sleep duration and academic performance.
- H_1 : Association exists

Observed frequency table:

Sleep duration/ academic performance	Excellent	Good	Average	Poor	Total
5-6 hrs (R1)	6	29	13	0	48
6-8 hrs (R2)	10	49	33	0	92
Less than 5 hrs (R3)	2	3	4	0	9
More than 8 hrs(R4)	2	9	4	2	17
Total	20	90	54	2	166

Observed (pooled) frequency table:

Sleep duration/academic performance	Excellent	Average	Good+poor	Total
5-6 hrs (R1)	6	13	29	48
6-8 hrs(R2)	10	33	49	92
R3+R4	4	8	12	26
Total	20	54	90	166

Expected frequency table:

	Excellent	Average	Good +poor
R1(n=48)	5.783	15.614	26.024
R2(n=92)	11.084	29.952	49.880
R3+R4(n=26)	3.133	8.434	14.096

$$\chi^2(\text{cal}) = 1.7917$$

$$\text{Degree of freedom} = (3-1)(3-1) = 4$$

$$\chi^2(\text{tab}) = 9.488$$

Result:

$$\chi^2 (\text{calculated}) < \chi^2 (\text{table})$$

Decision:

We fail to reject H_0 at 5% level of significance:

Interpretation:

The computed chi-square value of 1.7917 is less than the tabulated value of 9.488 at 4 degree of freedom and 0.05 level of significance.

Therefore, the null hypothesis is accepted- there is no significant association between sleep duration and academic performance.

This means that the number of hours a student sleeps does not significantly affect their academic performance based on the collected data.

Analysis 4: Correlation matrix and multiple correlation

R1.23- How academic performance depends on physical activity and mental well- being.

R2.13- How physical activity depends on academic performance and mental well- being.

R3.12- How mental well-being depends on physical activity and academic performance.

1. Academic performance

2. Physical activity

3. Mental well-being

Formula:

$$R = [(r_{12}^2 + r_{13}^2 - 2r_{12}r_{13}) / (1 - r_{23}^2)]^{1/2}$$

$$r_{12} = 0.135637$$

$$r_{13} = 0.098802$$

$$r_{23} = 0.056667$$

Correlation matrix:

1	0.136	0.099
0.136	1	0.057
0.099	0.057	1

Multiple correlation	Value
R1.23	0.1635
R2.13	0.1424
R3.12	0.108

Interpretation: The correlation matrix reveals the pairwise relationships among the three study variables. All correlations are positive but very weak in magnitude.

The correlation between academic performance and physical activity ($r_{12} = 0.136$) indicates a very weak positive association, suggesting that students who engage in physical activity shows a slight tendency toward better academic performance, but the relationship is not substantial. Similarly, the correlation between academic performance and mental well-being ($r_{13} = 0.099$) reflects a very weak positive link, implying that better mental well-being is marginally associated with improved academic outcomes. The correlation between physical activity and mental well-being ($r_{23} = 0.057$) is the weakest among all

three pairs, indicating that physical activity has a negligible linear relationship with mental well-being in the present sample.

Analysis 5: Simple linear regression analysis

Regression: Impact of physical activity on academic performance

This analysis examines whether impact of physical activity influences academic performance among college students.

$$Y = \beta_0 + \beta_1 X + \epsilon$$

Where:

- Y = Physical Activity
- X = Academic Performance
- β_1 = Regression Coefficient
- β_0 = Intercept

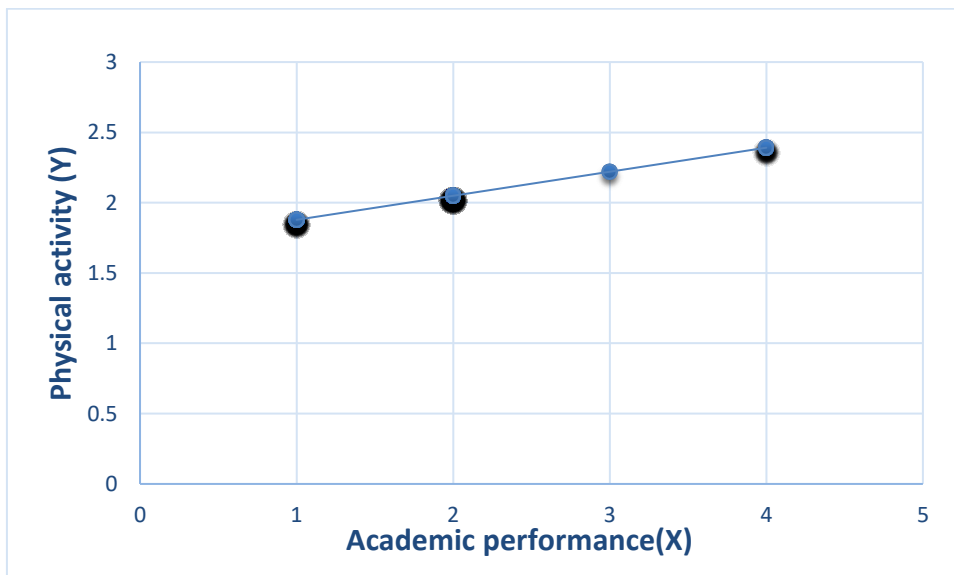
Estimated Regression Equation:

$$Y(\text{est}) = 1.7076 + 0.1709X$$

Regression Coefficient(β_1)= 0.1709

Intercept(β_0) = 1.7076

$$R^2 = 0.0184$$



Interpretation:

- The positive regression coefficient (0.1709) indicates that academic performance increases with physical activity. As students perform better academically, there is a slight tendency to engage more in physical activity.
- Specifically, a 1-unit increase in academic performance leads to a 0.1709 unit increase in physical activity level among students.
- The intercept value of 1.7076 indicates that when academic performance is zero, the baseline level of physical activity is 1.7076.
- The R^2 value of 0.0184 indicates that academic performance explains only 1.84% of the variation in physical activity. This is very low, suggesting that 98.16% of variation in physical activity is influenced by other factors such as personal motivation, time availability, lifestyle habits, and institutional facilities.
- The regression analysis confirms a very weak positive influence of academic performance on physical activity, which is consistent with the correlation coefficient ($r_{12} = 0.136$) obtained earlier in the study.

Conclusions:

1. Physical Activity Among Students

- A significant portion of respondents engage in physical activity occasionally or rarely, indicating that regular exercise habits are not well established among college students.
- Running/walking and sports are the most common forms of physical activity, while gym attendance remains less frequent.

2. Academic Performance

- Majority of respondents rate their academic performance as average to good.
- Students who study 2-4 hours daily form the largest group, suggesting moderate study engagement among the sample.

3. Mental Well-being

- A large proportion of Students report feeling stressed due to studies often or sometimes, indicating academic pressure is a common mental health concern.
- Most students agree or strongly agree that exercise helps reduce stress, showing awareness of the mental benefits of physical activity.

4. Relationship Between Physical Activity and Academic Performance

- A weak positive relationship exists between physical activity and academic performance ($r_{12} = 0.136$).
- Students who are more physically active show a slight tendency toward better academic outcomes, but the relationship is not substantial.

5. Relationship Between Physical Activity and Mental Well-being

- The correlation between physical activity and mental well-being is the weakest among all pairs ($r_{23} = 0.057$).

- Despite students believing exercise reduces stress, actual behavioral data shows a negligible linear relationship between exercise habits and mental well-being scores.

6. Correlation and Multiple Correlation Insight

- All three variables show very weak positive relationships with each other.
- Academic performance has the relatively strongest combined association with physical activity and mental well-being ($R_{1.23} = 0.1635$), making it the most connected variable among the three.

7. Regression Result (Key Statistical Evidence)

- A positive relationship exists between:
- Academic Performance $\rightarrow$ Physical Activity
- Specifically, a 1-unit increase in academic performance leads to a 0.1709 unit increase in physical activity level among students.
- However, academic performance explains only 1.84% ($R^2 = 0.0184$) of the variation in physical activity, confirming that other factors such as personal motivation, lifestyle, and time availability play a much larger role.

Suggestions:

1.Promote Physical Activity in Colleges

- Sports and fitness activities should be made compulsory at college level.
- Colleges should provide proper gym, sports ground, and fitness facilities.
- Regular sports events and competitions should be organized to encourage participation.

2. Reduce Academic Stress

- College schedules should be designed to balance study and physical activity time.
- Students should be encouraged to take short exercise breaks during long study sessions.
- Counselling sessions should be provided to help students manage academic pressure.

3. Improve Mental Well-being

- Mental health awareness programs should be conducted regularly in colleges.
- Students should be educated about the connection between exercise and mental health.
- Yoga and meditation sessions should be introduced as part of college routine.

4. Encourage Healthy Lifestyle

- Students should be motivated to maintain a regular sleep schedule of 6-8 hours daily.
- Healthy eating habits should be promoted by reducing fast food consumption on campus.
- Water intake and nutrition awareness programs should be conducted.

5. Strengthen Academic Support

- Teachers should adopt interactive and less stressful teaching methods.
- Extra coaching and academic support should be provided to average and poor performing students.
- Study groups and peer learning should be encouraged among students.

6. Create Awareness About Benefits of Exercise

- Seminars and workshops on physical and mental health benefits of exercise should be held.
- Success stories of physically active students should be shared to motivate others.
- Social media should be used to spread awareness about healthy student lifestyle.

7. Role of College Administration

- College administration should develop a structured physical activity policy.
- Dedicated time slots for physical activity should be included in the timetable.
- Student well-being committees should be formed to monitor health and academic balance.

8. Further Research

- Future studies should include additional variables such as sleep quality, nutrition, socioeconomic background, and peer influence.
- Larger and more diverse samples should be used for stronger and more generalizable findings.
- Longitudinal studies should be conducted to understand long-term effects of physical activity on academic performance and mental well-being.

Overall Conclusion:

The present study was conducted to examine the relationship between physical activity, academic performance, and mental well-being among college students. Data was collected from 166 students through a structured questionnaire covering various aspects of their exercise habits, academic outcomes, and mental health status.

The findings of the study reveal that although all three variables — academic performance, physical activity, and mental well-being — show positive relationships with each other, the associations are very weak in magnitude. This suggests that these three constructs largely operate independently among college students in the present sample.

A significant observation from the study is that despite students personally believing that exercise reduces stress and improves academic performance, their actual behavioral data does not strongly support this belief statistically. This gap between perception and behavior indicates that awareness alone is not sufficient — consistent practice of physical activity needs to be encouraged and institutionally supported.

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Appendix:

1. Gender *

- ☐ Male
- ☐ Female
- ☐ Other

2. Age (in years) *

- ☐ 17-19
- ☐ 20-22
- ☐ 23-25
- ☐ Above 25

3. Course *

- ☐ Undergraduate
- ☐ Postgraduate
- ☐ PhD
- ☐ Other

4. Institute/Faculty/Department in which you are enrolled? *

- ☐ Science
- ☐ Social Science
- ☐ Arts
- ☐ Commerce
- ☐ Laws
- ☐ Medical

5. Academic Year? *

- ☐ 1st Year
- ☐ 2nd Year
- ☐ 3rd Year
- ☐ 4th Year
- ☐ Other

6. Residence *

- ☐ Campus Hostel
- ☐ Home
- ☐ PG/Rented

- Other

PHYSICAL ACTIVITY SECTION:

7. How many hours do you study daily? *

- Less than 2 hours
- 2-4 hours
- 4-6 hours
- More than 6 hours

8. How would you rate your academic performance? *

- Excellent
- Good
- Average
- Poor

9. Do you perform physical activity regularly? *

- Yes
- No
- Sometimes

10. How often do you exercise in a week? *

- Daily
- 3-5 days
- 1-2 days
- Rarely

11. What type of physical activity do you usually do? *

- Gym
- Running/Walking
- Yoga
- Sports

12. How long do you exercise daily? *

- Less than 30 minutes
- 30-60 minutes
- 1-2 hours
- More than 2 hours

13. Do you participate in sports activities? *

- ☐ Regularly
- ☐ Occasionally
- ☐ Rarely
- ☐ Never

14. Mode of transport to the college? *

- ☐ Cycle
- ☐ Bike/ Scooter
- ☐ Walk
- ☐ College Bus
- ☐ Other

15. What motivates you to exercise? *

- ☐ Health Benefits
- ☐ Stress Relief
- ☐ Fitness/Appearance
- ☐ Doctor or Family Advice

16. What prevents you from exercising regularly? *

- ☐ Lack of Time
- ☐ Academic Pressure
- ☐ Lack of Interest
- ☐ Health Issues

17. Do you feel energetic after physical activity? *

- ☐ Strongly Agree
- ☐ Agree
- ☐ Neutral
- ☐ Disagree

MENTAL WELLBEING SECTION:

18. Do you feel stressed because of studies? *

- ☐ Always
- ☐ Often
- ☐ Sometimes
- ☐ Rarely

19. Does exercise help reduce your stress? *

- ☐ Strongly Agree
- ☐ Agree
- ☐ Neutral
- ☐ Disagree

20. How many hours do you sleep daily? *

- ☐ Less than 5 hours
- ☐ 5 to 6 hours
- ☐ 6 to 8 hours
- ☐ More than 8 hours

21. Do you think exercise helps to manage exam stress? *

- ☐ Yes
- ☐ No
- ☐ Maybe
- ☐ Not Sure

22. Do physically active students perform better academically? *

- ☐ Yes
- ☐ No
- ☐ Maybe
- ☐ Not Sure

23. Does physical activity increase your study motivation? *

- ☐ Strongly Agree
- ☐ Agree
- ☐ Disagree
- ☐ Neutral

24. How often do you eat fast food? *

- ☐ Daily
- ☐ 2-3 times a week
- ☐ Once a week
- ☐ Rarely

25. How do you feel after having fast food? *

- ☐ Energetic
- ☐ Normal

- Sleepy
- Guilty

26. Your daily water intake? *

- Less than 1 litre
- 1-2 litres
- 2-3 litres
- More than 3 litres

27. How often do you visit the health centre? *

- Frequently
- Occasionally
- Rarely
- Never

OPINION BASED QUESTIONS:

28. Colleges should encourage sports and fitness activities? *

- Strongly Disagree
- Disagree
- Neutral
- Agree

29. Students who exercise regularly are mentally healthier? *

- Strongly Disagree
- Disagree
- Neutral
- Agree
- Strongly Agree

30. Overall, physical activity improves students' academic and mental wellbeing? *

- Strongly Agree
- Agree
- Neutral
- Disagree